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FEATURES OF CROP MATERIALS

Features of crop materials impact agricultural practices, storage, and marketing. They determine the behaviour of the product for its handling and also play important role in design of conveying equipment, storage structures, separation equipment etc. These unique features could be broadly classified into five.

1. Physical properties (shape, size, density, colour, texture, appearance, surface area, volume, porosity)
2. Biological nature (living organism, moisture content)
3. Perishability
4. Seasonal supply
5. The need for processing before consumption: Most crops require some form of processing before they can be consumed or utilized. This can include cleaning, sorting, drying, milling, or other transformations to prepare them for market.

Density

Density is a measure of mass per unit volume of materials and it is measured in g/cm^3 or kg/m^3**.** The density of grains decides the size of screening surface. Density is also essential for determination of flow behaviour, separation processes and packaging requirement of crop materials. The density of materials is not constant and changes with temperature (higher temperatures reduce the density of materials) and pressure. Density of a material can be calculated as:

$$\boldsymbol{Density = \frac{Mass\ of\ the\ material\ (kg)}{Volume\ of\ the\ material\ (m^3)}}$$

The mass of a material can be determined using a top-pan balance.

Volume of regular shaped material: A ruler can be used to measure the length, breadth and height, and the volume can be calculated as:

Volume = length × breadth × height

Volume of Irregular-shaped material: Volume can be determined using a measuring cylinder or a displacement can depending on the quantity of the material.

Determination of the volume by Platform scale: Platform scale is a simple technique that is commonly used to determine the volume of large materials like fruits and vegetables. The weight of material is determined by weighing on the scale in air. Then, the material is forced into the water with the help of a rod and the weight of the displaced water is taken by difference. The volume is then calculated as:

$$\boldsymbol{Volume\ (m^3) = \frac{Weight\ of\ displaced\ water\ (kg)}{Density\ of\ water\ (kg/m^3)}}$$

Note: Density of water is 1 g/cm³ or 1000 kg/m³

Some terminologies relating to density are listed below.

- **i. Particle/ Unit/True density:** The density of the solid material itself, excluding air spaces.
- **ii. Bulk density:** The density of the material including air spaces between particles.
- **iii. Apparent density:** The density of a porous material, including the volume of pores.

SPECIFIC GRAVITY

The density of liquids can be expressed as *specific gravity* (SG), which is calculated by dividing the mass (or density) of a liquid by the mass (or density) of an equal volume of pure water at the same temperature:

Specific Gravity = mass of liquid / mass water

Specific Gravity = density of liquid / density of water

If the specific gravity of a liquid is known at a particular temperature, its density can be found using:

Liquid density = Specific gravity × density of water

MOISTURE CONTENT

Moisture content is the amount of water present in a material. Crop materials usually have high moisture content at harvest and this make them susceptible to microbial growth and spoilage. Moisture content affect other properties of a material and also influence the postharvest handling of such material. There are different methods of measuring moisture content of a material. However, oven drying method is the most commonly used. In this method, a known weight of a material is dried in an oven at a particular temperature for a specific time until the final weight is constant. Moisture content is then calculated on wet or dry basis as:

$$\boldsymbol{Moisture\ content\ on\ wet\ basis}\ (\%), MC_{wb} = \frac{W_I - W_f}{W_I} \times \mathbf{100}$$

$$\boldsymbol{Moisture\ content\ on\ dry\ basis}, MC_{db} = \frac{W_I - W_f}{W_f} \times \mathbf{100}$$

Where: W_i is the initial weight of the product before drying while W_f is the final weight of the product after drying.

VISUAL PROPERTIES OF CROP MATERIALS

Visual properties are a subset of physical properties that are directly observable by sight. They help in assessing the overall quality and maturity of crops. They aid farmers and processors in making informed decisions about handling, processing and marketing their crops. Examples include shape, size, colour, texture and appearance.

Importance of visual assessment

1. Equipment Design: Different sizes and shapes can affect storage volume, separation efficiency, and even the design of equipment and processing machinery like harvesters, planters, conveyors and sorting machines. For instance, the size of grains determines the mesh size of sieves used in cleaning and separation processes; and shape of fruits and vegetables dictates the design of conveyors and sorting machines. Visual properties like size and shape are also crucial in the design of appropriate storage structures.
2. Storage: Visual characteristics play a role in determining storage methods and duration. For instance, the color and appearance of fruits and vegetables can be used to assess their ripeness and storage potential.
3. Sorting, Separation and Quality Assessment: Color is a key indicator of maturity, ripeness and quality in many crops. It can be used to automate sorting and grading processes, ensuring uniform product quality. For example, ripe fruits and vegetables will have a specific color, and changes in color can signal spoilage or overripeness. Color also influences consumer perception and marketability of crops. Appearance encompasses surface texture and overall visual appeal, which are also important for marketability and consumer acceptance of crop materials. Visual inspection and color analysis are common methods for assessing crop quality.
4. Handling and Transportation: Visual properties like size and shape affect how easily crops can be handled, transported, and stored. Understanding these properties is crucial for optimizing storage space and preventing damage during handling.

SEPARATION OF CROP PRODUCE

Separation of crop materials encompasses various techniques to isolate desired components like seeds or grains from harvested crops, removing unwanted materials like chaff, weeds or broken pieces. The separation processes ensure the purity and quality of the harvested crop for further use or processing. Common separation methods include threshing, cleaning,

Methods of crop separation

1. **THRESHING**: This is the separation of grains from the stalks, usually by trampling on, beating or shaking the harvested material. Machines such as threshing machine and combine harvester which can harvest, thresh and winnow the grains on the field are now being used for this operation. Threshing is accomplished by impact action in most conventional threshers. A rotating cylinder (spike-tooth cylinder) and a concave screen are utilized to accomplish threshing. As the cylinder rotate, the seed is detached by impact force from its panicle and drops on a reciprocating screen which separates the straws from the seed. Types of threshers include wheat, sorghum, millet guinea corn threshers and so on. The components of most threshers comprises of the following: a. Hopper: to hold the agricultural materials before threshing b. Spike-tooth cylinder: to detach the seed from its panicle. c. Concave/Screen: to separate seeds from undesirable materials d. Chute: to discharge processed agricultural materials.

2. CLEANING OF CROP MATERIALS

Cleaning is the unit operation which involves the removal of contaminating materials such as soil, plant debris, metals, chemical residues, oil etc. from crop materials in a suitable condition for further processing. Cleaning of crop materials should be done at earliest opportunity to prevent damage to processing equipment, and to save time and money. Cleaning process can be categorised into two:

A. Wet cleaning: Wet cleaning is more effective than dry methods for removing soil from root crops or dust and pesticide residues from fruits and vegetables. It is also dust less and causes less damage to crops and equipment than dry methods. Different combination of detergent and sterilant at different temperatures allow flexibility in wet cleaning operations. However the use of warm cleaning water may accelerate chemical and microbiological spoilage. In addition, wet procedures produce large volumes of effluent, with high concentration of dissolved and suspended solids. Hence, procurement of clean water and disposal of the effluent are also required. Re-circulated, filtered and treated water can be re-used whenever it is possible, to reduce cost. Wet cleaning equipment include spray washers, drum washers, brush washers and flotation tank.

B. Dry Cleaning: Dry cleaning procedures are used for products that are smaller, have greater mechanical strength and possess lower moisture content (grains and nuts). Separation by dry procedures generally involves smaller and cheaper equipment than wet procedures and produces a concentrated dry affluent which may be disposed off more cheaply. The main groups of equipment used for dry cleaning include air classifiers, magnetic and electrostatic separators, physical separators and separators based on the screening of foods and imaging machines.

i. Air classifiers: These use a fast moving stream of air to separate contaminants from foods using the difference in the densities and the projected areas of particle. Air classifiers have very wide application for removing solids from lighter solids and air or liquids from vapour. Air classifiers are widely used in harvesting machines to separate heavy contaminants and light contaminants from grain. Winnowers and aspirators are examples of air classifiers. The air velocity required is calculated as:

$$Vc = \sqrt{[\frac{4d\ (\rho_s - \rho)}{\mathbf{3}C_d\rho}]}$$

Where: Vc (ms^{-1}) is the minimum velocity, C_d is the drag coefficient, ρ_s is density of the solid and ρ is density of the fluid.

Winnowing involves the use of wind or forced air to separate lighter materials like chaff from grains. This process helps in the removal of weevil or other pest in the grains. It can be done manually by dropping the grain and chaff from a height and letting the wind blow away the lighter chaff while the heavier grains fall backdown for recovery. Hand-held winnowing baskets can be used to shake the seeds to separate out the dirt and chaff. Other techniques including the use of Fanning mills or a winnowing fan (a shaped basket shaken to raise the chaff) or using a tool (a winnowing fork or shovel) on a pile of harvested grains.

i. Magnetic and Electrostatic Separators: This method involves the use of magnet to remove metallic components from crop produce. Magnetic cleaning involves cascading the contaminated product over one or more magnates, using magnetic drums, magnetised conveyor belts or magnets located above conveyors. Both permanent and electrical magnet can be used. Nevertheless, electrical magnets are easier to clean (by simply switching the power off) while permanent magnet cleaners are cheaper.

ii. Screening: screens are size separators which separate contaminants of different size from that of the raw material. Industrial screens can be rotary drum screens, flat bed screens and reciprocating screens.

iii. Physical separators: Separation of contaminants from crop material is possible when the crop has a regular well defined shape such as peas and blackcurrant seeds. Such seeds are separated from contaminants by exploiting their ability to roll down on an inclined conveyor belt. A disk separator consists of a series of vertical metal discs with precisely engineered indentations in the sides. It is used to separate grain seed from weed seeds. The indentation match the shape of the grain and as the disc rotates the grain is lifted out and removed.

3. SORTING

Sorting is the separation of crops into categories on the basis of a measurable physical property such as size, shape, colour, weight and density, to obtain crops with uniform qualities. Like cleaning, sorting should be employed as early as possible to ensure a uniform product for subsequent processing.

Shape and size sorting: This involves the separation of crop materials into two or more fractions on the basis of differences in size and shape. It is particularly important if the food is to be cooled or heated as the rate of heat transfer is in part determined by the size of the individual piece and variation on the individual pieces can cause over processing or under processing. Shape and size sorting is achieved by belt and roller sorter for fruits, and by disk sorter and screen sorters such as drum screen sorter for tubers.

Colour sorting (Photometric sorting): Foods may be sorted at a very high rate using microprocessor controlled colour sorting equipment. As crop materials are passing through an illumination, a photo-detector measure the reflected colour of each piece and compare it with a preset standard. Crop materials are separated by a short blast of compressed air. In another system, arrays of pulsed laser beams are used to illuminate products as they are discharged from a conveyor. Reflected light is measured by a microprocessor, which operates an automatic reject system.

Weight sorting: This method is more accurate than other method and it is used for more valuable food materials like fruits. A typical weight sorter consists of a slated conveyor which transports the product over a series of counterbalanced arms. The conveyor operates intermittently and while stationary, the arms rise and weigh the fruits. Heavy fruits are discharged into the padded chute and light fruits are replaced on the conveyor to travel to the next weigher.

Aspiration and flotation sorter use differences in density. Grains pulses and nuts are sorted by aspiration. Green beans and peas are sorted by flotation in brine. The dense mature pieces sink while the younger pieces float.

4. GRADING

Grading, also known as quality separation, is the assessment of a number of attributes to obtain an indication of overall quality of food. Sorting on the basis of one attribute may be used as part of grading operations but not vice versa. Grading standards include size and shape, maturity, texture, flavour and aroma, freedom from blemish and contamination. Grading is carried out by trained operators who are able to assess a number of factors simultaneously. The graders form a balance judgment of the overall quality and physically separate the crop materials into quality categories.

CROP PROCESSING MACHINES

1. **Shelling Machines**: Shelling machines are machines which apply mechanical force in removal of kernel from its casing (or enclosure) in order to free it from any attachment or confinement e.g. removal of palm kernel from its nuts, removal of melon and moringa kernels from their shells. The principle of operation of shelling machines depends on the nature of the agricultural materials. For instance most motorized shellers use impact force in removing kernels from their encasement. This involves the application of mechanism such as flat bar or rasps bar cylinder with concave screen. In this case the seeds are loaded into the hopper from where it moves into the shelling chamber. It is then beaten by the rotating cylinder to free the kernels from their casing. During this process, the kernels are being separated from their host and pass through the screen to be discharged at the chute. It should be noted that the principle of shelling depends relatively on the engineering properties of biomaterials to be shelled, as it varies from one type to another. Other shelling machines include melon sheller, palm nut cracker, bush mango seed shelling machine etc.
2. **Milling Machines**: Milling is the physical process of reducing the size of solid materials like grains, by grinding, crushing or cutting them into smaller particles like flour or meal. This crucial unit operation involves cleaning, conditioning, and then mechanically grinding raw ingredients. The choice of mill depends on the raw material and the scale of production. There are two main types of milling.

i. Dry milling: This involves reducing the size of dry crops like wheat and corn, to produce products such as flour, meal and grits.
ii. Wet milling: This involves the processing of crop materials in a liquid medium such as water, to break them down into finer particles. Dry milling offers advantages like better heat dissipation and reduced dust.

There are a large number of mills available for specific crop materials and some common ones are as follows:

A. **Hammer mills** are almost universally used throughout the developing world. Small-scale hammer mills range in size from 2kW to 20kW. The hammer mill consists of a rotor carrying a collar that bears a number of hammers around its periphery which rotates inside a circular or rectangular rotor. A breaker plate is used to enhance communition, while a screen is used to recycle material bigger than the required size of product. Size reduction is mainly due to impact force. The size of final product can be varied by varying rotor speed, feed rate, clearance between hammers and grinding plate, number and size of hammers and the size of the screen. The milled grain is filtered out through a perforated plate that runs around the edge of the mill chamber. The mill is good for hard crystalline solids, fibrous materials and is used extensively for poultry feed manufacture. Grain for human food is ground to a 1mm particle size while animal food is ground to a 3 mm particle size. The advantages include versatility and freedom from damage by foreign materials. The disadvantages of the mill are high power requirement, non-uniform grinding and it may be unsuitable for wet milling.

The above mills can be operated in three modes:

I. Free flow of materials through the mill in a single pass
II. The exit from the mill is restricted by a screen and food remains in the mill until the particles are sufficiently small to pass through the screen apertures (under these 'choke' conditions, shearing forces play a larger part in the size reduction)
III. Recirculation through the mill of all material or larger pieces until sufficient size reduction has been achieved

B. **Disc attrition mills**: This mill consists of a set of two hard surfaced circular plates pressed together and rotating with relative motion. The material is reduced as it passes between the two plates. Different size of feed is accomplished by varying the pressure on the plates using spring or hand screw. Disc mills can be single disc, double disc and burr mills. In single disc mill feed passes between a high speed rotating grooved disc and the stationary casing of the mill. Intense shearing action result in comminution of the feed. In double disc mill the casing contain two rotating discs, rotating in opposite directions giving a greater degree of shear. Such attrition mills are widely used in cereal preparation, corn and rice milling. The burr mill is an older type of attrition mill, originally used in flour milling. Two circular stones are mounted on a vertical axis. The upper stone which is often fixed has a feed entry port while the lower stone rotates. Feed material passes to the gap between the upper and the lower stone. In some modes both stones rotate in opposite direction. Such mills are still in use in food industry and different variations of the mill exist. There are manual versions of the plate mill available, though they are arduous and require hard work to use.

C. **Ball mills:** These have a slowly rotating, horizontal steel cylinder which is half filled with steel balls 2.5–15 cm in diameter. At low speeds or when small balls are used, shearing forces predominate. With larger balls or at higher speeds, impact forces become more important. This combination of impact and shearing forces brings about a very effective size reduction. Ball sizes are usually in the range of 25 to 150 mm. Small balls give more point contact while larger balls give greater impact. They are used to produce fine powders, such as food colourants. A modification of the ball mill named a *rod mill* has rods instead of balls to overcome problems associated with the balls sticking in adhesive foods.

D. **Roller mills:** Roller mills are widely used to mill wheat. Two or more steel rollers revolve towards each other and pull particles of food through the 'nip' (the space between the rollers). The main force is compression but, if the rollers are rotated at different speeds, or if the rollers are fluted, additional shearing forces are exerted on the food. The size of the nip is adjustable for different foods and overload springs protect against accidental damage from metal or stones.

3. **Decorticator**: A decorticator is a machine that is used in the removal of outer coat of any agricultural materials usually pods. The process of decortication is synonymous to shelling. The word decorticate comes from the Latin verb decorticare derived by adding the prefix de (meaning to be rid off) to the noun Cortex. The cortex refers to the outer layer of a botanical or anatomical object, often specifically to a layer which lies just beneath the very outer surface, and to general features such as bark, husk, shell or rind. Conventionally, decortication encompasses dehulling, husking, cracking, splitting, popping and shelling. Decortication process is more generic than shelling process. In this case a decorticator could be used to remove the bark of a stem, seeds pod etc. A decorticating machine is a very versatile machine. Most decorticators apply shear and impact force to bring about decortications. This process is one of the most delicate operations in crop processing because the structure of the seed must not be tampered with or deformed. Types of decorticators include cowpea, groundnut and jatropha fruit decorticators.

HANDLING OF CROP PRODUCE

Efficient crops handling is 'the organised movement of crop materials in the correct quantities, to and from the correct place, accomplished with a minimum of time, labour, wastage and expenditure, and with maximum safety'. Some advantages of efficient crop materials handling include:

1. Reduction in storage and operating space requirement.
2. Better stock control

3. Improved working conditions through bulk materials handling.
4. It lowers the cost of production.
5. It brings about less wastage of materials and operator time.
6. There is lower risk of accidents.

List of handling equipment

1. Conveyors
2. Elevators
3. Cranes and hoists
4. Trucks
5. Pneumatic equipment
6. Water flumes
7. Carts
8. Chutes
9. Tractor

CONVEYORS AND ELEVATORS

Conveyors are widely used in agro-processing industries for the movement of solid materials, both within unit operations, between operations and for inspection of foods. There are a large number of conveyor designs, produced to meet specific applications, but all types can only cover a fixed path of operation. In general, conveyors and elevators are best suited to high-volume movement, where the direction of flow of materials is fixed and relatively constant. They can also be used as a reservoir of work-in-progress. Common types include:

Belt conveyors: This consists of an endless belt which is held under tension between two rollers, one of which is driven. The belts may be stainless steel mesh or wire, synthetic rubber, or a composite material made of canvass, steel and polyurethane or polyester. Belt conveyors are the most economical powered conveyor and they are used for conveying products over long distances, at high speeds, or for incline/ decline applications. Flat belts are used to carry packed foods, and trough-shaped belts are used for loose materials. Flat belts may be inclined up to 45º, if they are fitted with cross slats or raised chevrons to prevent the product from slipping.

Oscillating conveyor: An oscillating conveyor moves crop produce through an upward and forward oscillating motion of a metallic trough or tube which is typically supported by inclined, articulating arms or legs. A motor drives an eccentric shaft or unbalanced motor, creating the reciprocating movement that gently propels crops along the trough's surface, minimizing mechanical damage to fragile produce like wheat or barley. Advantages of oscillating conveyors include gentle handling, low maintenance, and efficient, residue-free emptying of materials.

Gravity Conveyor: Gravity conveyor moves crop produce using a slight downward incline, where the natural force of gravity pulls the items over a series of unpowered rollers or skate wheels. The mechanism relies on a simple-framework of rollers anchored in a frame, allowing them to spin and facilitate the smooth movement of crops without any external power source. This cost-effective and low-maintenance design enhances efficiency in handling various harvested crops by providing a continuous flow and reducing manual labor.

Auger or Screw conveyors comprise a rotating helical screw inside a metal trough. They are available in a wide range of lengths and capacities, and they are usually powered by an electric motor. Long augers may be mounted on wheels for easy transportation. They are used to move bulk foods such as flour and sugar, or small-particulate foods including peas and grain. The main advantages are the uniform, easily controlled flowrate, the compact cross-section without a return conveyor and total enclosure to prevent contamination. They may be horizontal or inclined, but are generally limited to a maximum length of 6 m as above this, high friction forces result in excessive power consumption.

Chain conveyors: These are used to move churns, barrels, crates and similar bulk containers by placing them directly over a driven chain, which has protruding lugs at floor level. It uses a powered continuous chain, typically made of interconnected links, that rotates around sprockets to transport heavy loads and unit loads like pallets and industrial containers. The motor drives the sprockets, which grasp the chain, creating a continuous loop that moves along guide rails. The chain can also have attachments like pendant hangers, to carry specific products.

Pneumatic conveyors: These consist of a system of pipes through which powders or small particulate foods, such as salt, flour, peas, sugar, milk powder, coffee beans, etc., are suspended in recirculated air and transported at up to 20–35 m/s. The air velocity is critical; if it is too low, the solids settle out and block the pipe, whereas if it is too high, there is a risk of abrasion damage to the internal pipe surfaces. There are different types of pneumatic conveying with different ways of introducing materials into the pneumatic conveyor and removing them after conveying. Generation of static electricity by movement of foods is a potential problem that could result in a dust explosion when conveying powders, and is prevented by earthing the equipment, venting, or explosion containment and suppression techniques. Pneumatic conveyors cannot be overloaded, have few moving parts, low maintenance costs and require only a vacuum pump or a supply of compressed air at around 700×10^3 Pa.

There are many designs of *elevator*, but two common types are bucket and magnetic elevators:

1. **Bucket elevators**: They consist of metal or plastic buckets fixed between two endless chains. They have a high capacity for moving free-flowing powders and particulate foods. The shape and spacing of the buckets and the speed of the conveyor (15–100 m/min) control the flow rate of materials.
2. **Magnetic elevators** are used for conveying cans within canneries. They have a positive action to hold the cans in place, and are thus able to invert empty cans for washing, and they produce minimal noise.

Cranes: Cranes handle crop produce by using their primary mechanisms of hoisting, traveling, slewing or rotating and luffing to lift, move and position harvested crops or planting materials. A hoist raises the load, a travel mechanism moves the crane or trolley horizontally, slewing rotates the boom of the crane for broader reach, and a luffing mechanism adjusts the angle of the boom to change the working radius. These movements are powered by electric motors or hydraulic systems and they enable the crane to efficiently transport crops from fields to collection points, facilitate planting and move materials in the processing facilities.

Carts: Carts handle crop produce by providing a durable platform, large wheels for mobility on tough terrain, and a towed or self-propelled mechanism to move heavy loads efficiently from the field to storage or processing areas. Common types include simple hand-pushed trolleys and tractor-towed grain carts or trailers, which can feature tilting mechanisms and large-capacity cargo beds. The selection of a cart depends on the produce, terrain, and scale of operation. Use of cart reduces handling, minimize losses and speed up logistics.

Trucks: Trucks handle crop produce by providing large capacity, fast and efficient transport using various mechanisms like refrigerated units and canvas curtains to control temperature and elements and employing specific loading, bracing and stowing practices to prevent damage, shifting and spoilage during transit. It encompasses temperature control to maintain freshness, physical bracing and cushioning to immobilize produce and protective measures such as canvas curtains to shield against sun and weather.

Applications of crop materials-handling equipment

	Conveyors	Elevators	Cranes and hoists	Trucks
Direction				
Vertical up		*	*	
Vertical down		*	*	
Incline up	*	*		
Incline down	*	*		
Horizontal	*			*
Frequency				
Continuous	*	*		
Intermittent			*	*
Location served				
Point	*	*		
Path	*			
Limited area			*	
Unlimited area				*
Height				
Overhead	*	*	*	
Working height	*			*
Floor level	*		*	*
Underfloor	*			
Materials				
Packed	*	*	*	*
Bulk	*	*	*	*
Solid	*	*	*	*
Liquid				*
Service				
Permanent	*	*	*	
Temporary			*	*

CALCULATING THE CAPACITIES OF CONVEYORS AND THE COST OF CONVEYANCE OF CROP MATERIALS

For belt conveyor:
Capacity, Q = V × W × D × T
Where Q = Capacity (tonnes/hr), V = Belt speed (m/s), W = Belt width (m), D = Bulk density (Kg/m^3), T = A factor accounting for fill level and troughing angle

Example
A belt conveyor has a width of 0.5 meter and a belt speed of 1 m/s. If the bulk density of the grains is 700 kg/m^3 and the fill factor is 0.5, calculate the capacity of the conveyor.

Solution
Capacity, $Q = V \times W \times D \times T$
$Q = 1\ m/s \times 0.5\ m \times 700\ kg/m^3 \times 0.5$
$Q = 175\ kg/s$
Converting to tonnes/hr
$Q = 175\ kg/s \times 60\ s/min \times 60\ min/hr \times 1000\ kg/tonne$
$Q = 630\ tonnes/hr$

For screw conveyor

Capacity, $Q = (\pi/4) \times (D^2 - d^2) \times p \times n \times D \times A$

Where Q = Capacity (m^3/hr), D = Screw diameter, d = Shaft diameter, p = Screw pitch, n = Revolutions per minute, D = Bulk density, T = Filled Area Factor (Accounts for how much of the screw's volume is actually filled with material, typically 0.45 to 0.65).

Example.
Calculate the capacity of a screw conveyor with the following information.
Screw diameter, D = 0.3 m, Shaft diameter, d = 0.1 m, Screw pitch, p = 0.2 m, Revolutions per minute, n = 100 rpm, Bulk density of crop produce = 700 kg/m^3 (0.7 tonnes/m^3), Filled area factor is 0.55.

Solution
Capacity, $Q = (\pi/4) \times (D^2 - d^2) \times p \times n \times D \times T$
$Q = (3.14/4) \times (0.3^2 - 0.1^2)) \times 0.2 \times 100 \times 0.7 \times 0.55$
$Q = 0.785 \times 0.08 \times 0.2 \times 100 \times 0.7 \times 0.55$
$Q = 45.8$ tonnes/hr.

$$\text{Cost of conveyance} = \frac{\text{Total capital cost + Total operating cost}}{\text{Total volume of crop conveyed}}$$

Capital cost: The initial investment in the conveyor system and any land and installation.
Operating cost: Energy cost, labour cost, tolls/taxes
Lifecycle costs: The total cost of the conveyor system over its entire lifespan, including all upfront and operational expenses.

Example
Calculate the cost of conveyance if the total initial investment and operational costs to move 10,000 tons of corn from field to a storage facility 5 km away is ₦50,000,000.

Solution

$$\text{Cost of conveyance} = \frac{\text{Total capital cost + Total operating cost}}{\text{Total volume of crop conveyed}}$$

$$\text{Cost of conveyance} = \frac{\text{₦50,000,000}}{\text{10,000 tonnes}}$$

Cost of conveyance = ₦5,000 per ton

DRYING

Drying can be defined as the application of heat to remove the majority of the water normally present in a food by evaporation. It involves the simultaneous application of heat and removal of moisture from foods, resulting in the process of mass and energy transfer.

Drying Methods:

Depending on the mode of heat and mass transfer there are four major methods of drying: conduction, convection radiation and freeze drying.

<u>Conduction</u>: Conduction is the movement of heat by direct transfer of molecular energy within solids e.g through metal containers or solid foods. Applicable dryers operate by subjecting the product to heat by direct contact with a heated surface. In this case, the dryer can be a pan or some other container in which the product is kept. Heat is supplied below the pan. To avoid burning, such dryers are equipped with stirrer for turning the product.

<u>Convection</u>: Air is heated and forced through the product which is usually enclosed in a container such as a storage bin, silo, drying chamber, conventional oven. *Convection* is the transfer of heat by groups of molecules that move as a result of differences in density (for example in heated air) or as a result of agitation (for example in stirred liquids). In the majority of applications all three types of heat transfer occur simultaneously but one type may be more important than others in particular applications.

<u>Radiation</u>: *Radiation* is the transfer of heat by electromagnetic waves. Drying is achieved by the application of energy from a radiating source to the product. Microwave ovens and solar energy dryers fell in this category.

<u>Freeze dryers</u>: In this method, moisture in the product is frozen by refrigeration. It is then sublimed into vapour by the application of heat under low pressure.

The different types of dryers available can be grouped into three: namely batch dryer, deep dryers and continuous dryers.

The main purpose of dehydration is to extend the shelf life of foods by a reduction in water activity. This inhibits microbial growth and enzyme activity. Therefore any increase in moisture content during storage, for example due to faulty packaging, will result in rapid spoilage. The reduction in weight and bulk of food reduces transport and storage costs. There are two basic mechanisms involved in the drying process of crop. First is the migration of moisture from the the interior of each crop to the produce and the second is the evaporation of moisture from the surface to the surrounding air. The two major types of drying rates associated with drying of agricultural materials are:

1. Constant drying rate: Drying commences and water moves from the interior of the food at the same rate as it evaporates from the surface, the surface remains wet. This is known as the *constant-rate period* and continues until a certain *critical moisture content* is reached. This is mostly the case for highly perishable products like fruits and vegetables.

2. Falling drying rate: The falling-rate period is usually the longest part of a drying operation. Rate of moisture removal is decreasing because all free surface water and water of saturation are removed.

Importance and purpose of drying crop produce

1. Drying extends the shelf life of crop produce by inhibiting microbial growth and enzyme activity.
2. It makes crop produce easier to handle.
3. It reduces volume and weight of produce by removing moisture, making them less expensive to store and transport.
4. It improves product stability

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Parameters for drying

There are a large number of factors that control the rate at which foods dry, which can be grouped into the following categories:

I. Those related to the processing conditions
II. Those related to the nature of the food
III. Those related to the drier design.

Major drying parameters

1. Air temperature
2. Air velocity
3. Humidity
4. Drying time.
5. The properties of the materials being dried (moisture content, thickness, composition and heat sensitivity).

Components of a drying system

1. Air heating system (heater, heat exchanger, combustion chamber) to supply heat.
2. Air circulation system (Fan, air distribution, airflow adjustments, duct) to move the heat and channel the heated air.
3. Material handling system (loading/ unloading system, drying chamber/ bin, tray, racks).
4. Control and monitoring (Temperature sensors, humidity sensors, control panel/ PLC safety features) to regulate drying parameters.
5. Insulator to ensure minimal heat loss.
6. Other components include dust collection system, cooling system, moisture meters and air filtration system.

STORAGE PESTS AND DISEASES

A wide range of biological factors influence stored products. Losses may be qualitative or quantitative, or a combination of both, and result from the inability of the host to limit biological damage. Produce with surface blemishes, such as discoloration, blotches or signs of insect activity, may be rejected by graders or consumers and hence give rise to qualitative losses. Quantitative losses arise when stored produce is directly consumed by primary insect pests and rodents, or from the rapid and extensive decay caused by the action of micro-organisms.

A stored product pest can be defined as ‘any organism injurious to stored foodstuffs of all types (especially grains, pulses and fruits), seeds, and diverse types of plant and animal materials’. These pests can be categorised in various ways, for example, on the basis of systematics, dependence on stored products, feeding habits and damage caused, commodity infested, ecological niches, time taken to reach economically important levels, the value losses they cause or the type of handling or manufacturing industry in which they are found.

Physical and economic damage of storage pests and diseases

1. They can cause physical damage of the crops.
2. They cause reduction in viability of stored produce.
3. They reduce the quantity of produce by consuming it and then contaminate it through their faeces.
4. They cause deterioration in the quality of the produce.
5. They can cause damage to the storage infrastructure.
6. Spot of injuries by insects may predispose crops to disease attack.
7. They increase the cost of production during the cause of controlling them.
8. The profits of farmers are reduced.
9. They render vegetables and fruits unattractive and unmarketable.
10. They may cause consumers’ heath deterioration.

PREVENTION MEASURES AGAINST RODENTS IN STORED PRODUCTS

1. Maintaining proper sanitation and hygiene including effective cleaning and proper waste management practices.
2. Physical control measure such as fencing round the farm with wire nets.
3. Practicing proper storage techniques (Use air-tight containers, off the floor storage and FIFO)
4. Structural and environmental controls (seal entry points, secure doors and control the environment).
5. Inspection and monitoring: Carry out regular inspection of the store, of any incoming shipments and monitor the traps regularly).

CONTROL MEASURES AGAINST RODENTS IN STORED PRODUCTS

1. Physical control such as setting of traps to catch rodents and shooting rodents with gun.
2. Biological control which involves the introduction of natural enemies of rodents to keep their population under control.
3. Chemical control such as the use of rodenticides
4. Integrated control which involves the use of two or more of the above method. This is more effective.

PROCESSES OF DETECTING INSECTS IN STORE

Traditional and direct methods

1. Visual inspection

2. Insect traps: Pheromone traps use food-based attractants to lure and capture insects. Sticky traps capture crawling and flying insects, light traps employ specific wavelengths of light to attract and capture insects.
3. Sampling and sieving

Advanced and Indirect methods

1. Monitoring devices: Acoustic sensors detect the sound of insect activity like movement and feeding. Electronic noses identify volatile compounds and infrared sensors detect heat emitted by insects.
2. Product analysis: X-ray imaging scan detects hidden insects or damage.

STORAGE PESTS

1. Insects and arachnids e.g beetle, moths, weevils and termites
2. Rodents e.g rats
3. Birds e.g house sparrow
4. Fungi (moulds and yeasts)
5. Bacteria

TYPES OF INSECTICIDES USED IN STORE

1. Chemical or synthetic organic insecticides
   i. Organophosphates e.g Malathion, Pirimiphos-methyl, Diazinon and Dichlorvos.
   ii. Pyrethroids e.g Deltamethrin, Resmethrin and Permethrin
2. Inorganic insecticides such as Aluminium Phosphide, Carbamates and Neonicotinoids
3. Biological insecticides such as *Bacillus thuringiensis*
4. Botanical insecticides such as Azadirachtin.

CONTROL OF INSECTS IN STORED PRODUCTS

1. Traditional Techniques: This encompasses sanitation and hygiene and physical methods such as handpicking of insects and larva, making physical barriers, use of Berlese funnel, carrying out manual inspection and setting traps.
2. Modern techniques: These techniques include the use of insecticides (chemical methods), Temperature and moisture control, Aeration and atmosphere control, Biological control, Botanical extracts, Pheromones, Radiation, Ozonation and insect growth regulators.

MICROBIOLOGICAL ORGANISMS CAUSING STORAGE LOSSES

1. Fungi (Mould and yeasts): Fungi that grow under conditions of low oxygen tension are adapted to growing in cereals stored under an airtight regime. These include *Paecilomyces variotii*, *Scopulariopsis candida*, *P. roqueforti*, *Candida* spp, *Byssochlamys fulva* and *B. nivea*. Patulin is the predominant mycotoxin expected in this case, produced by *P. variotii* and *Byssochlamys* spp. *Aspergillus parasiticus* is the most important postharvest fungus on groundnuts, with about 70% of isolates taken from groundnuts able to produce aflatoxins. Fungi that invade tree nuts include *Penicillium commune*, *P. crustosum*, *P. solitum*, *P. funiculosum*, *P. oxalicum*, *P. citrinum*, *Aspergillus flavus*, *A. wentii* and *A. versicolor*. Mycotoxins include cyclopiazonic acid, citrinin, penitrem A, aflatoxin and sterigmatocystin. Major fungi of stored perishable produce are species of *Fusarium*

(teleomorphs *Gibberella* and *Nectria*), *Phytophthora, Alternaria* (teleomorph *Lewia*), *Rhizopus, Pythium, Sclerotinia, Botrytis* (teleomorph *Botryotinia*), *Penicillium* (teleomorphs *Eupenicillium* and *Talaromyces*), *Cladosporium* (teleomorphs *Mycosphaerella* and *Venturia*), *Phoma* (teleomorph *Pleospora*) and *Rhizoctonia* (teleomorphs *Thanatephora*, *Helicobasidium*).

Yeast gain entry through wounds and may produce ethanol as they grow, so that infected produce gives off an odour of fermentation. Some examples are *Saccharomyces* on strawberries, *Candida* on carrots and yams, *C. krusei* on citrus, *Nematospora* on beans and *Kluyveromyces marxianus* on onion.

2. Bacteria: Important post-harvest bacteria are in the genera *Pseudomonas, Xanthomonas* and *Erwinia* (though some species of *Erwinia* have recently been moved to the genus *Pectobacterium*).

3. Viruses: Viruses are not normally of post-harvest significance but may detract from the market value of the produce. They enter plants through wounds, from sap-sucking insects, mechanical abrasion or harvesting activity. Viruses are mainly disease agents of field crops and they are not often encountered in storage, though there are a few examples, such as spraing (brown lines in the flesh of infected potato tubers). Spraing is caused by tobacco rattle virus or potato mop-top virus. Infection by cauliflower mosaic virus and turnip yellows virus leads to mosaic and yellowing of Cruciferae, and cucumber mosaic virus is sometimes found on cucumber fruits, varying in degree from a slight pale mottling to the development of white distorted areas. Other viruses of post-harvest interest are internal cork in sweet potato (caused by sweet potato feathery mottle virus), internal brown spot of yams (caused by dioscorea badnavirus), apple ringspot (from infection by apple chlorotic leafspot virus and apple stem pitting virus) and stony pit of pears (also caused by apple stem pitting virus).

4. Nematodes: Nematodes are not major pests of stored produce, but can cause some problems on sweet potatoes and yams if they are brought into the store on infested tubers. The sweet potato nematode is *Meloidogyne incognita*, which causes cracking, shrinkage necrosis and occasional hyperplasia in storage. Stored yams are attacked by *Pratylenchus coffeae*, which causes cracking of the skin and gives the tuber a corky appearance.

CONTROL OF MICROBIOLOGICAL ORGANISMS IN STORE PRODUCTS

1. Physical methods: These methods include Temperature control like freezing and refrigeration, packaging like canning, drying, radiation and filtration.
2. Chemical methods such as the use of preservatives and controlling the pH.
3. Hygiene and sanitation
4. Monitoring

STORAGE AND PRESERVATION OF CROPS

Agricultural products are usually seasonal. Therefore for them to be available through the year, they must be stored for a given period of time. The agents of spoilage include enzymes, environmental conditions (heat, moisture) insects and rodents, microorganisms and physiological processes (respiration, transpiration and sprouting). The main function of a crop storage structure is to control the activities of these agents. Crop storage is an important aspect of post harvest technology.

Storage can be defined as the art of keeping the quality of agricultural materials and preventing them from deterioration for specific period of time, beyond their normal shelf life.

PARAMETERS FOR SAFE STORAGE

Moisture content, relative humidity and storage temperature are the three major parameters that must be monitored and controlled during storage to prevent excessive storage losses. Moisture content is the amount of water molecule contained in an agricultural material. In storage, of more importance is the equilibrium moisture content. This is because the stored product interacts with its immediate environment since agricultural material is hygroscopic. When products are kept in an environment, there is a continuous interaction of moisture between the product and the environment. At a point, the product attains equilibrium with the environment and the moisture interaction ceases. This is called the Equilibrium Moisture Content (EMC). The EMC of any stored product is affected by climatic factors such as rainfall, relative humidity and temperature. Regions with high relative humidity tend to keep crop at a high EMC and this is not safe for storage. The interaction between EMC, relative humidity and temperature is called moisture isotherm. High temperatures lead to moisture migration and moisture condensation in stores.

Other parameters that could be considered for safe storage include physical protection of the products, effective pest control, hygiene and segregation.

PHYSIOLOGICAL FACTORS AFFECTING CROP STORAGE AND QUALITY

Postharvest loss can be defined as the degradation in both quantity and *quality* of a food production from harvest to consumption.

The major physiological factors affecting crop storage and quality are respiration, transpiration and sprouting. Other factors include ripening, senescence, and ethylene production. These factors significantly affect post-harvest crop quality and storage life by degrading stored produce. Managing these processes through controlled temperature, humidity, and atmosphere helps extend shelf life and maintain desirable attributes like texture, color, and flavor. High moisture content, for example, promotes respiration, fungal growth, and insect activity, making it a critical factor in determining safe storage conditions.

Respiration: This metabolic process consumes stored reserves, releasing heat, water, and carbon dioxide, and is directly influenced by temperature, with rates roughly doubling for every 10°C rise. High respiration rates accelerate deterioration and quality loss.

Transpiration: This is the process of water loss from crops, leading to wilting and loss of turgor. Maintaining high humidity levels (around 90-95%) in storage environments is crucial to minimize water loss and prevent wilting.

Sprouting: This is a major physiological factor that significantly decreases crop quality and storability by consuming stored energy, increasing metabolic activity, and leading to dry matter loss. This process breaks crop dormancy, resulting in reduced shelf life, decreased nutritional value, and changes in desirable characteristics like texture and flavor. Factors such as temperature, humidity, harvest time, and varietal characteristics influence when sprouting occurs, emphasizing the need for controlled storage conditions to prevent or delay it

Ethylene Production: Ethylene is a plant hormone that promotes ripening and senescence. High temperatures can increase ethylene production, which, in turn, accelerates the aging and deterioration of many fruits and vegetables.
Ripening: This involves a series of metabolic changes that occur after harvest, leading to changes in color, texture, and flavor. While desired at consumption, uncontrolled ripening during storage reduces shelf life.
Senescence: This is the natural aging process in plants and their parts, leading to irreversible deterioration. Delaying senescence extends the storage life of harvested crops.
Enzymatic Activity: Post-harvest enzymatic activity contributes to the breakdown of tissues, leading to softening, loss of color, and changes in flavor, all of which degrade quality

Storage and preservation methods for perishable foods

To prevent spoilage and ensure safety, perishable foods require specific storage and preservation methods:

- **Refrigeration:** Keeping perishable foods at temperatures between 0°C and 4°C slows down the growth of spoilage-causing microorganisms and extends shelf life.
- **Freezing:** Freezing at -18°C or lower halts microbial activity and preserves food for several months. However, it is important to note that freezing can affect the texture and quality of some crops including fruits and vegetables.
- **Vacuum sealing:** Removing air from packaging reduces the growth of aerobic bacteria and extends the shelf life of perishable foods.
- **Use of preservatives:** Adding natural or chemical preservatives can inhibit microbial growth and extend the shelf life of perishable foods.
- **Drying (Dehydration)**: Removing water content inhibits microbial growth by denying them the moisture they need.
- **Fermentation:** Using beneficial microorganisms to preserve food, which also increases the shelf life.
- **Canning:** Heat is used to destroy microbes, and the food is sealed in airtight containers to prevent recontamination.
- **Vacuum Packing:** Removing air from packaging reduces the growth of aerobic bacteria, extending shelf life.

Storage and preservation methods for non-perishable foods

While non-perishable foods are inherently stable, proper storage is still important to maintain their quality:

- **Cool, dry storage:** Like semi-perishable foods, non-perishable items should be stored in a cool, dry place to prevent moisture absorption and spoilage.
- **Airtight containers:** Using airtight containers helps protect non-perishable foods from moisture, pests, and contamination. Domestic storage can use sealed plastic, metal, or

glass containers, as well as jute sacs and bags, to minimize oxygen and retard pest activity.

- **Rotation:** Practicing the "first in, first out" method ensures that older items are used before newer ones, reducing the risk of spoilage.
- **Proper sealing:** Ensuring that packaging is properly sealed prevents exposure to air and contaminants, preserving the quality of non-perishable foods.
- **Silos and Bins**: Bulk grain is stored in large, enclosed containers like silos, which protect it from moisture, pests, and contamination.
- **Modified Atmosphere Storage**: Reducing the oxygen content and increasing carbon dioxide in the storage environment can prevent spoilage.